Master's Dissertation

The scaling laws governing the dynamics of rigid, pressure-driven spheres in narrow, fluid-filled elastic tubes

Author: Sajid Ahmed

Supervisor: Prof. Anne Juel

In fulfilment of the requirements for the degree of Integrated Master's of Physics

in the

Manchester Centre for Nonlinear Dynamics

Department of Physics and Astronomy

The University of Manchester

June 2025

Acknowledgements: The experimental work for this project was conducted in collaboration with Thomas Sigsworth under the supervision of Professor Anne Juel at the University of Manchester.

Abstract

This report describes the experimental investigation of the power laws governing the motion of rigid spheres driven by hydraulic pressure through narrow, lubricated elastic tubes. The theoretical motivation of this research was provided by the work of Rallabandi et al.[1], who derived three distinct scaling regimes for the force-velocity relationship of a rigid axisymmetric intruder in a lubricated elastic tube. The objective of this investigation was to test these scaling laws experimentally.

The importance of studying this system lies in its similarities to many real-world phenomena, including the motion of red blood cells in capillaries, the movement of eggs in oviducts, soft robotics, and industrial manufacturing processes that rely on material extrusion.

We conducted our experiments using a microfluidic pump to apply hydraulic pressure to the rigid spheres (metal balls) inside a tube connecting two fluid reservoirs filled with rapeseed oil. High-framerate videos of the experiment were recorded and analysed using a custom-built Python program to extract the average intruder velocities and perform linear regression fits to determine the power-law exponents.

The experiments yielded results that strongly indicate the presence of the intermediate-speed power law $\mathcal{F} \propto V^{2/3}$ for the 18 mm ball and high-speed power law $\mathcal{F} \propto V^{4/5}$ for the 11 mm, 13 mm, 14 mm, and 15 mm balls. Similar exponents were identified for these intruders in the non-dimensional λ parameter space.

These results strongly support the existence of the intermediate- and high-speed power laws and lend empirical support to the theoretical treatment by Rallabandi et al.[1]. In future work, experiments should include ellipsoid intruders and aim to eliminate axial stretching. This would provide a more reliable comparison with the theory in λ parameter space.

Contents

Abstract 1

1 Introduction **5**

2 Theory **7**

2.1 Low speed regime 9

2.2 Intermediate speed regime 10

2.3 High speed regime 10

3 Methodology **11**

3.1 Experimental procedure 12

3.2 Velocimetry 13

3.3 Piecewise power law analysis 14

3.4 Oil viscosity 14

3.5 Young's modulus 14

4 Results **16**

4.1 Pressure-velocity scaling 16

4.1.1 10 mm 16

4.1.2 11 mm 17

4.1.3 12 mm 18

4.1.4 13 mm 19

4.1.5 14 mm 20

4.1.6 15 mm 21

4.1.7 18 mm 22

4.2 Dimensionless scaling 23

5 Discussion **25**

5.1 Methodological limitations 25

5.2 Comparison in λ parameter space 26

5.3 Future work 27

6 Conclusion **29**

Bibliography **33**

Appendices **33**

A Table of Results **34**

B Images **35**

C Code **36**

C.1 Velocimetry 36

C.2 Piecewise Linear Regression 41

List of Figures

2.1 Schematic diagram of a spherical intruder in a narrow elastic tube. 7

3.1 Schematic diagram of the experimental apparatus. 11

4.1 Pressure-Velocity plot for 10 mm diameter ball. 16
4.2 Pressure-Velocity plot for 11 mm diameter ball. 17
4.3 Pressure-Velocity plot for 12 mm diameter ball. 18
4.4 Pressure-Velocity plot for 13 mm diameter ball. 19
4.5 Pressure-Velocity plot for 14 mm diameter ball. 20
4.6 Pressure-Velocity plot for 15 mm diameter ball. 21
4.7 Pressure-Velocity plot for 18 mm diameter ball. 22
4.8 Dimensionless pressure-lambda plot for balls through 11 mm to 18 mm. . 23

B.1 Position-time plot for 12 mm ball at 80 mbar pressure 35
B.2 A frame from the video of the 12 mm ball falling at 80 mbar pressure . . . 35

List of Equations

1.1 Predicted power laws 6
2.1 Lubrication thickness ODE 8
2.2 Dimensionless force 8
2.3 Maximum radial excess 8
2.4 Dimensionless speed parameter 8
2.5 Axial length scale for a sphere 9
2.6 Slender intruder condition 9
2.7 Dimensionless radial displacement 9
2.8 Low speed power law 9
2.9 Dimensionless curvature for a sphere 10
2.10 Intermediate speed power law 10
2.11 High speed power law 10
4.1 Thin film area 23
4.2 Dimensionless pressure 24
4.3 Hoop stress resistance 24

Chapter 1

Introduction

This report investigates the elastohydrodynamic problem of a rigid, axisymmetric intruder being driven through a narrow elastic tube filled with a lubricating fluid[1,2]. Specifically, we examine the power laws relating the propulsive force on the intruder to its velocity. Such a system is ubiquitous in natural and engineered settings, and a rigorous characterisation of its force-velocity relationship is crucial for diverse applications[1,2].

Elastohydrodynamic transport of slender bodies occurs in a variety of biological processes. For example, eggs traverse the oviduct[3], a soft, tubular channel, while lubricated by mucosal secretions in the reproductive systems of birds and reptiles[4]. In these systems, the propulsive force is often provided by the rhythmic contractions of the oviduct walls[5,6].

Similarly, the movement of a bolus of food through the oesophagus and intestines is an example of elastohydrodynamic transport, where the bolus is lubricated by saliva and mucus. A process of muscular contractions and spasms called peristalsis provides the propulsive force to move the bolus through the narrow, elastic digestive tracts[7].

Furthermore, red blood cells in blood must rely on the lubrication provided by blood plasma to squeeze through very narrow capillaries, which are scarcely wider than the red blood cell itself[8,9]. These capillaries then provide essential oxygen to tissues and organs such as the muscles, heart and brain, and are essential for the survival of the organism[10,11].

Technological applications include futuristic microfluidic drug-delivery devices, in which therapeutic particles or droplets navigate soft polymeric channels under precisely controlled pressures to deliver targeted drugs to specific sites within the body[12,13].

A better-known example of elastohydrodynamic transport in the medical field is coronary angioplasty. During this procedure, a rigid, compressed metal mesh called a stent is pushed through the blood vessel to the treatment site, and a balloon inside the stent is inflated to expand the mesh, which then supports the arterial walls to maintain blood flow through the vessel[14,15].

In manufacturing, modelling rigid intruders in elastic tubes optimises the extrusion of soft materials, where rigid dies shape compliant polymers. Understanding the elastohydrodynamic lift ensures smooth surfaces and prevents defects by maintaining a fluid film

between the die and the extrudate[16].

In soft robotics, this physics governs how rigid actuators or sensors move within fluid-filled, flexible chambers, as in the artificial muscle developed by Bridgestone in 2017[17]. Precise modelling of these interactions allows for predictable, biomimetic movement and prevents material fatigue[18].

Early theoretical treatments of lubricated motion in elastic tubes date to Lighthill's 1968 analysis[19]. He derived the square root scaling law between force and velocity $F \propto V^{1/2}$ by modelling a pressure-driven system that couples to the tube's hoop stress. While this analysis was elegant, it neglected the vitally important effects of axial stretching and membrane tension. Over time, experimental evidence for Lighthill's scaling law has been reported in numerous studies[10,11,20].

More comprehensive numerical studies by Rallabandi et al.[1,21] have since incorporated the effects of axial membrane tension into the partial differential equations governing the system, thereby developing a more comprehensive nonlinear membrane theory. Rallabandi, Eggers, Herrada and Stone solved the system of partial differential equations using direct numerical simulations in the 2021 paper 'Force on an axisymmetric intruder moving through a lubricated elastic tube'[1], and predicted three distinct scaling regimes:

$$F \propto \begin{cases} V^{1/2} & \text{low speed} \\ V^{2/3} & \text{intermediate speed} \\ V^{4/5} & \text{high speed} \end{cases} \tag{1.1}$$

where F is the drag force on the intruder, and V is the intruder velocity.

The low-speed power law is similar to the Lighthill analysis; however, the intermediate and high-speed power laws are novel predictions from the nonlinear membrane theory[1]. These state-of-the-art predictions enjoy broad theoretical consensus but lack extensive experimental validation. Therefore, the principal motivation of this report is to describe the experimental investigation of the force-velocity scaling of a rigid intruder in a lubricated elastic tube, and to assess the extent to which measured scaling laws corroborate the numerically derived ones.

Chapter 2

Theory

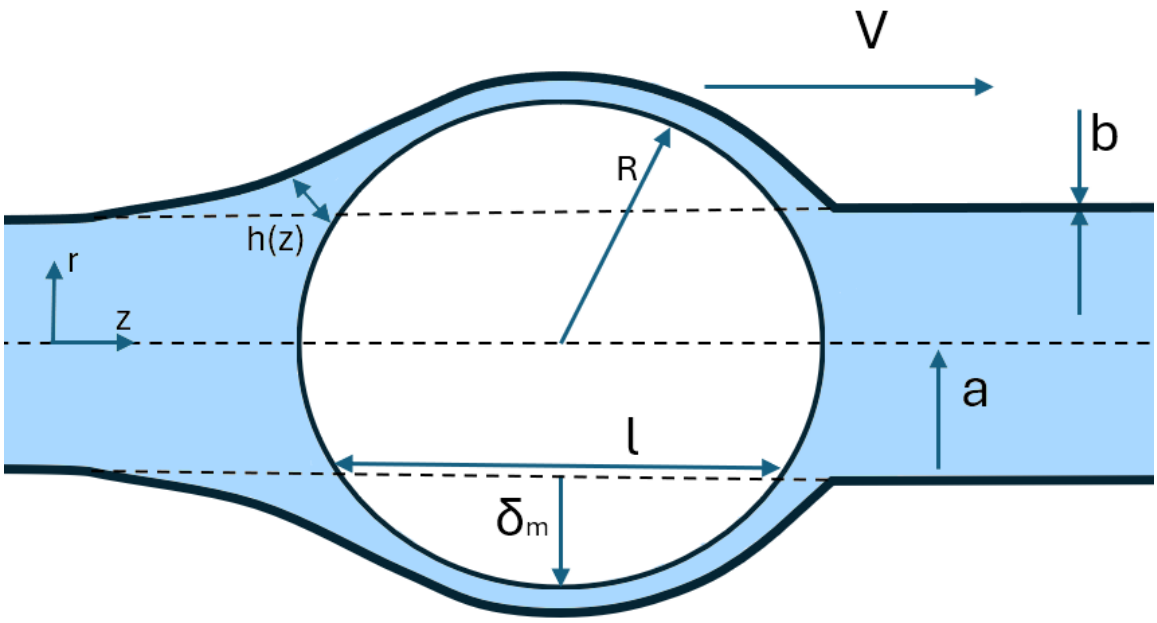

Figure 2.1: Schematic diagram of a spherical intruder in a narrow elastic tube.

Figure 2.1 shows a rigid, axisymmetric, and slender intruder moving at constant velocity through an elastic cylindrical tube filled with a viscous fluid. The intruder is slightly larger than the tube, which would lead to dry contact in the absence of motion, and the intruder would become stuck in the tube due to the hoop stress resistance[21]. However, as the intruder moves through the liquid-filled tube, a thin lubricating film of thickness $h(z)$ forms between the intruder and the tube wall, reducing friction[1].

The tube deforms outward by a radial excess related to the difference between the radii of the slender intruder and the undeformed tube, $\delta(z) = R(z) - a$. The assumption is made that the pressure loss effects of Poiseuille flow are negligible compared to the pressure of the lubrication film[1].

Through the coupling of the lubrication film pressure, hoop stress and axial membrane

tension, they derive an ordinary differential equation (ODE) for the lubrication film thickness:

$$\frac{dh}{dz} - \frac{6\mu V a^2}{Eb}\frac{(h-h^*)}{h^3} + \frac{d\delta}{dz} = 0 \tag{2.1}$$

where $h(z)$ is the fluid film thickness, $\delta(z)$ is the radial excess, μ is the dynamic viscosity of the fluid, V is the velocity of the intruder, a is the radius of the tube, E is the Young's modulus of the tube, b is the thickness of the tube wall, and $h^* \equiv \frac{2q}{V}$ is the effective gap thickness that can support the flux q in axial shear flow between locally cylindrical surfaces (determined as part of the ODE solution)[1].

They then rescale the equation in the limit of small velocities and use a method of perturbative asymptotic expansion to find analytical solutions, which are subsequently validated through direct numerical integration of the full nonlinear ODE[1].

When a slender, axisymmetric, rigid object moves through a rigid, fluid-filled tube, the thickness of the lubricating film is constant and independent of the object's velocity V. Therefore, the drag force on the object is expected to scale linearly with velocity as $F \propto V$, akin to a modified Stokes' flow[22].

However, for an intruder in a narrow elastic tube, Rallabandi et al. proved the film thickness scales as $h \propto V^{1/2}$ due to the coupling of the fluid pressure with the tube's elastic properties[1]. By identifying the $V^{1/2}$ scaling, the authors proved that at low speeds the elasticity of the tube reduces the drag force on an intruder due to the axial tension in the tube, which is caused by the friction between the intruder and tube[1].

The authors use a perturbative asymptotic expansion method to solve the ODE, and thus use non-dimensional parameters. They rescale the force on the intruder to acquire a dimensionless force using the scaling factor:

$$\mathcal{F} = \frac{2\pi E b \delta_m^2}{a} \tag{2.2}$$

where E is the Young's modulus (Pa), b is the tube wall thickness (m), a is the radius of the tube (m), and δ_m is the maximum radial excess (m), which for a spherical object is

$$\delta_m = R - a \tag{2.3}$$

where R is the radius of the sphere (m).

Furthermore, they define a dimensionless speed parameter as

$$\lambda = \frac{6\mu V a^2 L}{E b \delta_m^3} \tag{2.4}$$

where μ is the dynamic viscosity (Pa s), V is the velocity (m s^{-1}), and L is the axial length scale over which there would be solid contact between the intruder and the elastic tube (m) in the absence of a lubricant, which for a spherical object is

$$L = 2 \cdot \sqrt{R^2 - a^2} \tag{2.5}$$

These two dimensionless parameters form the basis of the scaling laws we will study in the following sections.

Furthermore, the authors define the condition for the intruder to be considered 'slender' as

$$s = \frac{\delta_m}{L} \ll 1 \tag{2.6}$$

This requires the maximum radial excess to be much smaller than the axial length scale over which the lubricating film acts[1].

Furthermore, the dimensionless radial displacement parameter is defined as

$$\epsilon = \frac{\delta_m}{a} \ll 1 \tag{2.7}$$

where a is the radius of the tube (m).

The definitions of these parameters are essential to understanding the subsequent analysis, because Rallabandi et al. perform their direct numerical simulations to study the dependence of the drag force $\mathcal{F}$ on the dimensionless speed λ and the dimensionless radial displacement ϵ[1].

2.1 Low speed regime

In the limit of very slow speed ($\lambda \ll 1$), the elastohydrodynamic drag on the intruder is found to obey

$$\mathcal{F} \approx \left(\frac{3}{4} S^0 \lambda\right)^{1/2} \tag{2.8}$$

where S^0 is the dimensionless curvature of the intruder at the equator, or the slope of the intruder's profile at the leading edge[1].

For a sphere, the dimensionless curvature at the equator is given by

$$S^0 = \frac{a}{R} \tag{2.9}$$

The theoretical analysis applies to intruders with arbitrary axisymmetric shapes, as the force depends only on the intruder's slope at the leading edge, where more gradual tapering substantially reduces resistance[1]. This scaling law is derived from an asymptotic analysis of the lubrication thickness ODE in the limit of small velocities and is corroborated by direct numerical simulations.

The fluid reduces friction between an oversized intruder and the tube walls by establishing an ultra-thin fluid film. The interplay of viscous shear in this film and hoop-stress elasticity gives rise to a power law, in which the tube's axial membrane tension dominates the drag force[1].

2.2 Intermediate speed regime

In the intermediate speed regime ($\lambda \geq 1$), both the viscous pressure in the gap and the hoop stress contribute comparably across the axial length scale. There is a larger fluid film than at low speeds, where the lubricating film thickness is determined by the competition between viscous drag pulling fluid into the gap and capillary pressure pushing it out[1]. In this regime, the drag on the intruder is dominated by the surface tension.

Direct numerical simulations give rise to the following scaling law for the drag force on the intruder:

$$\mathcal{F} \propto \lambda^{2/3} \tag{2.10}$$

Interestingly, this scaling law is analogous in form to the Bretherton scaling typically associated with air bubbles in rigid tubes[23].

2.3 High speed regime

In the high-speed regime ($\lambda \gg 1$), there is a three-way balance among the thin film lubrication pressure (which induces expansion in the tube), the hoop stress resistance (which resists this expansion), and the axial membrane tension (which distributes this lubrication pressure over an extended deformation length)[1].

The authors use direct numerical simulations to derive the following scaling law for the drag force on the intruder:

$$\mathcal{F} \propto \lambda^{4/5} \tag{2.11}$$

Chapter 3

Methodology

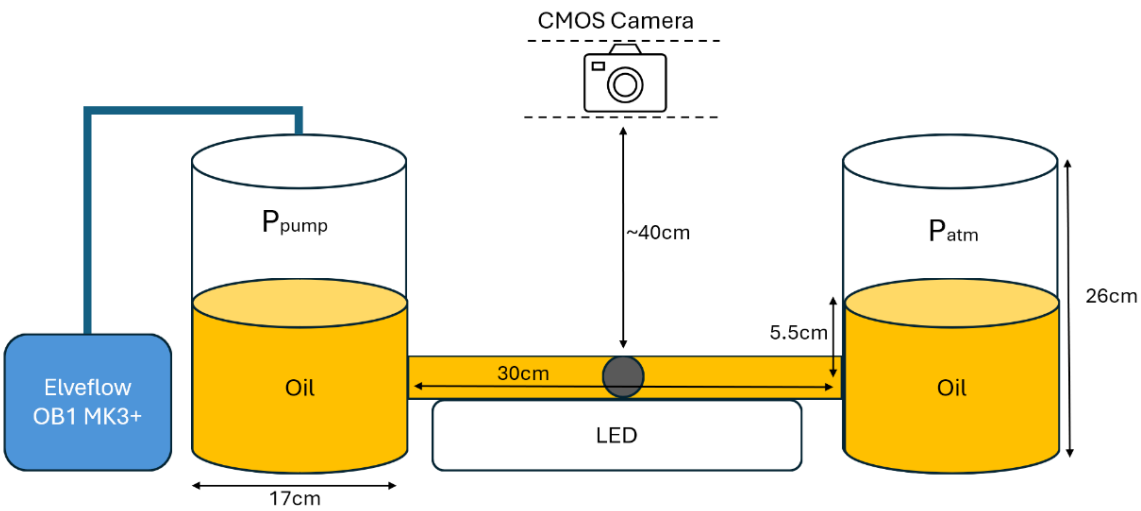

Figure 3.1: Schematic diagram of the experimental apparatus.

Figure 3.1 shows the tube connected at one end to a sealed and pressurised reservoir containing 1 L of rapeseed oil; the opposite end was connected to an open reservoir containing 1 L of rapeseed oil open to atmospheric pressure. The pressurised reservoir was sealed by solvent-welded polymer joints and reinforced with industrial-grade epoxy and a tightly wound clamp. The experiment was performed in an air-conditioned, temperature-controlled room in the Nonlinear Dynamics Laboratory at an ambient temperature of 20 ± 1 °C.

The slender intruders consisted of metal balls of diameters 10.00 ± 0.02 mm, 11.00 ± 0.02 mm, 12.00 ± 0.02 mm, 13.00 ± 0.02 mm, 14.00 ± 0.02 mm, 15.00 ± 0.02 mm and 18.00 ± 0.02 mm. The elastic pipe used was a silicone rubber tube of inner radius 4.99 ± 0.16 mm, wall thickness 0.57 ± 0.08 mm, and a length of 30 ± 1 cm, where the radius and thickness measurements were conducted by a PhD candidate under the supervision of Prof. Anne Juel in 2007[24]. It was produced in 2005 by Primasil Silicones Ltd[24]. The lubricant used was commercially available rapeseed oil with a measured dynamic viscosity

of 0.071 990 ± 0.003 413 Pa s at an ambient temperature of 20 ± 1 °C.

All experiments were performed using an Elveflow OB1 MK3+ microfluidic pump operated via the Elveflow Smart Interface v3.10[25]. The pump is specified to ± 0.1 mbar accuracy between 1 - 2000 mbar[26]. At the start of each experimental day, the pump was calibrated "dry" (disconnected from the tube) to ensure consistent pressure measurements across experiments.

Videos of the ball moving within the silicone tube were recorded using a high-framerate greyscale JAI industrial CMOS camera with a 6 mm focal-length ultra-wide-angle lens to capture the entire length of the tube, operated via the JAI eBUS SDK v6.5.1 software[27]. The camera was suspended directly above the tube using a custom-made rig secured to the desk to minimise movement and ensure the camera remained at a consistent height above the experimental apparatus. To maximise the framerate of the recordings, the region of interest was confined to the tube and its immediate surroundings ($\approx 150 \times 2000$ pixels), and all videos were saved as uncompressed TIFF file stacks.

These optimisations enabled us to achieve a maximum frame rate of 100 fps. However, the variability was considerable and may have been improved by using a camera with a modern USB-C connection that supports higher bitrate data transfer. The camera's exposure time was fixed at 300 µs to eliminate motion blur at higher translational ball speeds. A white LED backlight was mounted directly beneath the tube to maximise contrast against the ball and tube to optimise imaging. A sheet of lined paper (line spacing of 8 mm) was positioned between the LED and the tube for in-frame spatial calibration. The tube was positioned so that it lay completely horizontally on the LED platform to prevent the ball's position in the frames from being distorted or hydrostatic pressure uncertainties from being introduced.

3.1 Experimental procedure

The starting inlet pressure was 10 mbar for each ball and was increased in 5 mbar steps until the ball could overcome the membrane tensions and friction of the tube and thus move with a constant velocity along the entire length of the tube. This threshold was defined as the yield pressure, which increased with ball diameter due to greater tube deformation, leading to higher membrane tension.

Before each trial, the ball was held at the tube inlet while the pump reached the target pressure (stability within ±1 mbar). Once the pressure stabilised, the video recording was started, the ball was released, and its motion was captured until it reached the end of the tube. This procedure was repeated in increments of 5 mbar for $P < 100$ mbar, 10 mbar for 100 mbar $< P \leq 300$ mbar, and 20 mbar for $P > 300$ mbar. An upper pressure limit of 600 mbar was imposed due to recurring oil leakages at the tube-to-reservoir joints at pressures above 600 mbar.

The lined-paper grid provided an 8 mm calibration scale within each frame; the pixel-to-

distance conversion factor was estimated using ImageJ 1.54p software[28]. This calibration was repeated for every ball to account for minor shifts in camera or apparatus alignment.

3.2 Velocimetry

All video datasets were analysed using a custom-written Python script. The objective was to extract the ball's position in each frame, convert this to a physical displacement value, and then determine the average velocity via a linear regression of the displacement-time data.

Each trial's uncompressed TIFF stack was imported into the code, and the frames were converted to 8-bit greyscale. The colours were then inverted so that the ball appeared as a bright circular region on a dark background. The mean grey-value profile along each vertical pixel column was computed for each frame. The result was a one-dimensional intensity distribution whose most prominent peak corresponded to the horizontal position of the ball's centroid in the frame. Thus, the SciPy' find_peaks' function[29] was used to identify the highest peak in the distribution, and its centre position (in pixels) was recorded as the ball's centroid position.

Conveniently, the JAI camera software encoded timestamps into the TIFF filenames as hexadecimal numbers representing milliseconds since the software was loaded. Thus, a custom Python function was used to convert each hexadecimal timestamp to a decimal value in seconds. The TIFF stack enabled precise timestamping of consecutive frames, yielding a time series synchronised with the ball's position series that is much more accurate than using an estimate of time based on the camera's framerate.

The ball's physical displacement was plotted against time, and a linear least-squares fit was performed using NumPy's polyfit routine with degree 1[30]. The slope representing the average ball velocity and the standard error on the slope, derived from the covariance matrix, provided an estimate of systematic measurement uncertainty.

To verify that the ball achieved and maintained a constant velocity, residuals of the linear fit were examined. Deviations beyond the standard error indicated transient acceleration or measurement noise, so data segments exhibiting this non-uniform motion were excluded from the linear fit using a scikit-learn machine learning module called RANSAC (Random Sample Consensus)[31].

This analysis procedure yields highly accurate average speeds with associated uncertainty estimates for all pressure and ball-size combinations, facilitating direct comparisons across experimental conditions. The measurements produced by this code were compared with hundreds of manually measured velocities using ImageJ software by Thomas Sigsworth and found to agree.

3.3 Piecewise power law analysis

Furthermore, a custom Python script was written to identify and quantify power-law relationships between each ball's applied pressure and average velocity. Before fitting, each ball's characteristic yield pressure was subtracted from the measured pressures to ensure an accurate power law analysis. Because yield pressure could only be determined within a 5 mbar increment, it was taken as the midpoint between the first pressure value that exhibited uniform motion and the pressure value preceding it.

To locate contiguous regions obeying a single power law, each dataset was transformed to a base 10 logarithm scale, and the 'pwlf' (piecewise linear fitting)[32] Python module was used to scan all possible segment boundaries, seeking regions exhibiting statistically distinct gradients, which were treated as separate power law regimes. To avoid overfitting, the maximum number of fit segments was limited to 3 (determined by trial and error), and only power-law regimes spanning at least one order of magnitude in velocity were considered.

No reliable multiple-regime behaviour was detected within individual ball datasets; thus, each dataset was ultimately characterised by a single power-law across its valid pressure range.

3.4 Oil viscosity

The rapeseed oil's dynamic viscosity was determined via stress-relaxation experiments performed on an Anton Paar MCR series bulk rheometer[33]. 1 mL samples of oil were subjected to a constant strain, and the resulting stress decay was recorded as a function of time. Stress-strain plots exhibited linear behaviour, confirming that the oil is a Newtonian fluid at ambient temperature.

Viscosity was determined from the slope of the stress-strain rate plot. To account for ambient temperature fluctuations (lab temperature 20 ± 1 °C), measurements were performed thrice at temperatures of 19 °C, 20 °C and 21 °C. The mean viscosity at 20 °C was taken as the working value, with the standard deviation of all nine measurements providing an estimate of uncertainties due to random temperature fluctuations.

3.5 Young's modulus

The elastic modulus of the silicone tubing was measured using an Instron Universal Testing System[34]. Six specimens of silicone tubing were cut into rectangular sections of known dimensions and tested: three dry and three wet, which had been submerged in the rapeseed oil for 48 hours. Wet samples were measured to assess the potential effect of oil absorption on the silicone's elastic modulus[35], as this would be a highly relevant source of systematic error in our experiments conducted over multiple days.

Each specimen was extended at a constant strain rate of 3 mm min^{-1}, and the stress-strain response was recorded. For each curve, the small-strain elastic modulus, E, was determined from the linear region between 2 % and 5 % strain; the initial 0 % to 1 % region was excluded to negate slack removal effects that occur at the beginning of the test (while the 'curled' tube material is stretched out long enough to become 'flat'). Dry specimens yielded an average small-strain Young's modulus of 2.070 ± 0.140 MPa, while wet specimens yielded an average small-strain Young's modulus of 1.9100 ± 0.0823 MPa.

A paired t-test confirmed that the difference was not statistically significant at the 95% confidence interval ($p = 0.311 > 0.05$), likely due to the limited sample size. Consequently, the overall Young's modulus used in subsequent dimensionless number calculations was taken as the mean average of all six measurements, 1.990 ± 0.136 MPa.

The dry and wet samples were taken from two different silicone tubes from the same batch (as the 'wet' tube was submerged in oil for 48 hours, it could not be used for subsequent dry testing). Although the tubes were from the same batch, they may have had slightly different material properties if extruded from different blocks of silicone. However, any differences are unlikely to be statistically significant due to the manufacturer's strict quality assurance (Primasil Silicones Ltd).

For future work, measuring a single tube specimen cut into dry and wet halves would eliminate potential inter-sample variability in the measurement of intrinsic material properties. Alternatively, the elastic modulus of the used tubes could be measured to provide directly relevant data for the specific samples used in the experiments.

Chapter 4

Results

4.1 Pressure-velocity scaling

4.1.1 10 mm

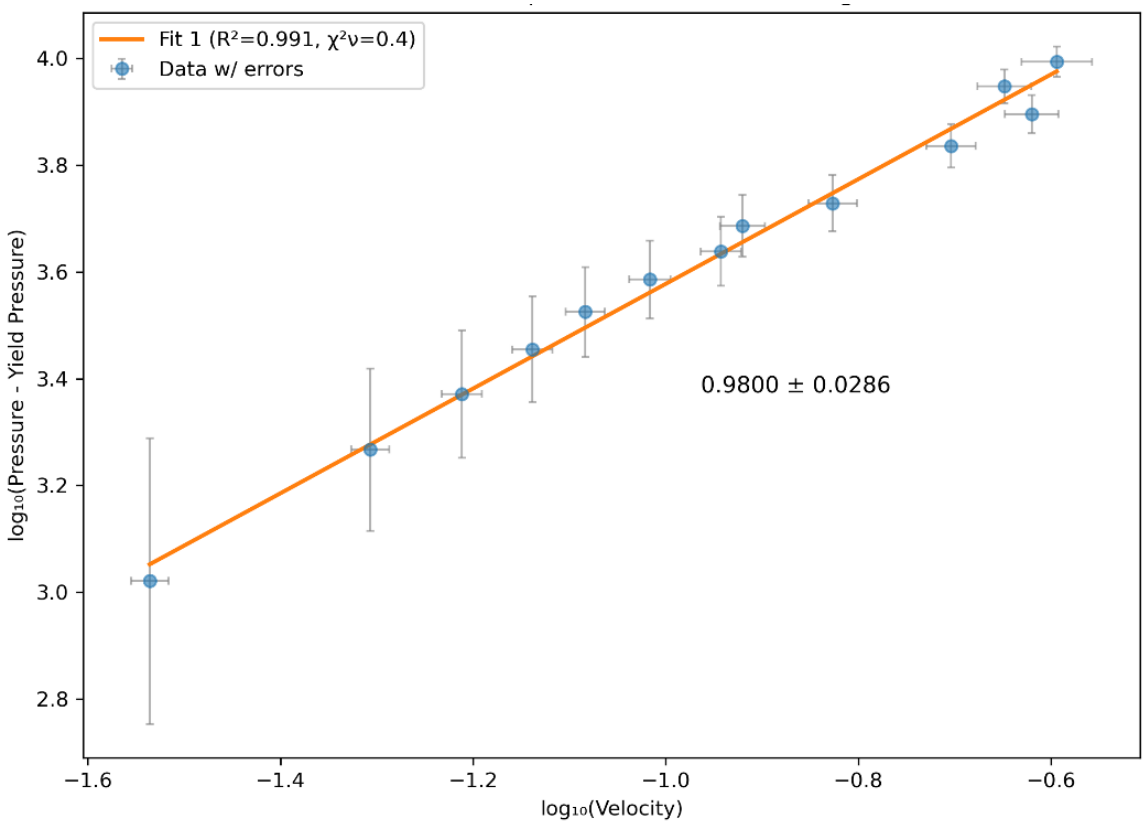

Figure 4.1: Pressure-Velocity plot for 10 mm diameter ball.

For the 10 mm ball, the value of the slenderness parameter is $s = 0.0158$, and radial excess parameter $\epsilon = 0.002$

The linear regression fit in Figure 4.1 indicates a power-law exponent of 0.9800 ± 0.0286

for the 10 mm ball over a pressure range of 1 kPa to 40 kPa, which corresponds to a linear scaling law between pressure and velocity. This observation is consistent with Couette flow dynamics. In the absence of tube deformation, the ball's motion is governed by oil flow in the tube, yielding a direct proportionality between pressure drop and velocity. In this zero-deformation regime, the motion is analogous to a ball in a rigid tube, with the lubrication film thickness being a constant and independent of translational velocity[36].

Although our measured maximum radial excess, $\delta_m = 0.01 \pm 0.161$ mm, nominally suggests a very small tube deformation, its magnitude is negligible compared to its uncertainty so that the true value may be close to zero. Furthermore, even if the ball does cause the tube to deform at low pressures, the tube's radial swelling at high pressures would exceed the small radial excess, essentially creating a zero-deformation regime at high pressures. Coupled with the experiment yielding a precise linear scaling, we are confident that any true excess is effectively zero, and that the nominal tube radius may have been slightly underestimated under working conditions. Since the 10 mm ball remains outside the deformation regime, the lubrication film pressure may be insufficient to render Poiseuille-flow pressure drops negligible, thereby contributing additional uncertainty to the measured exponent.

4.1.2 11 mm

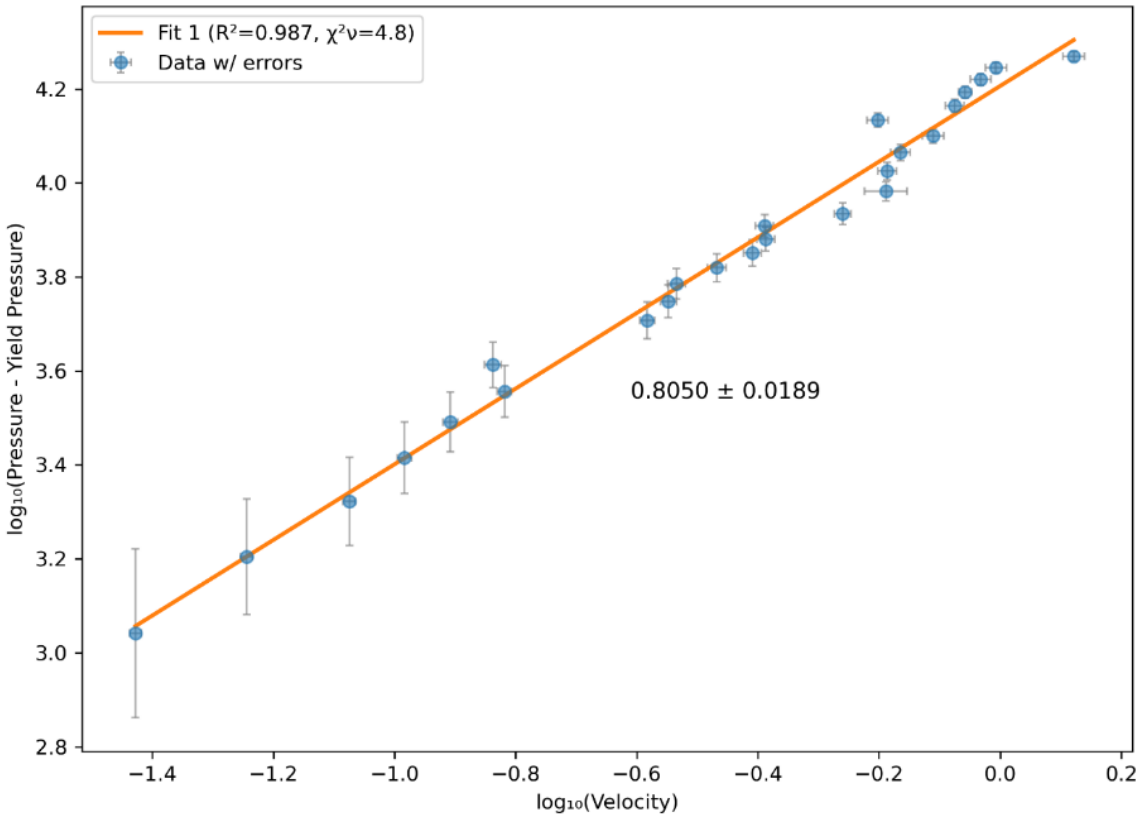

Figure 4.2: Pressure-Velocity plot for 11 mm diameter ball.

For the 11 mm ball, the value of the slenderness parameter is $s = 0.1102$ and radial excess parameter $\epsilon = 0.10$

Figure 4.2 indicates a power-law exponent of 0.8050 ± 0.0189 for the 11 mm ball over a pressure range of 1 kPa to 16 kPa, which corresponds to the high-speed $V^{4/5}$ (equation 2.11 power law[1].

This suggests that the experiment is in the regime of $\lambda \gg 1$, where a three-way balance exists among thin-film lubrication pressure, hoop stress resistance, and axial membrane tension. This leads to the film thickness scaling as $h \propto V^{2/5}$.

4.1.3 12 mm

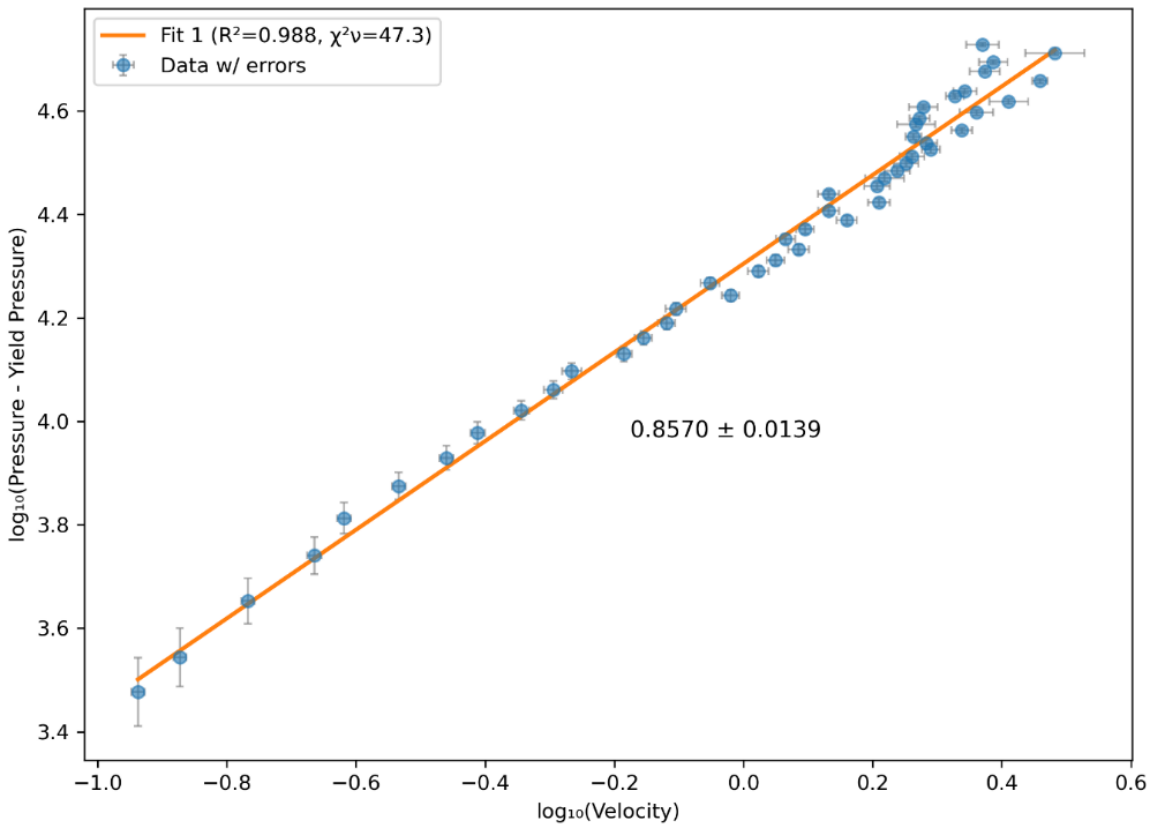

Figure 4.3: Pressure-Velocity plot for 12 mm diameter ball.

For the 12 mm ball, the value of the slenderness parameter is $s = 0.1516$ and radial excess parameter $\epsilon = 0.20$

Figure 4.3 indicates a power-law exponent of 0.8570 ± 0.0139 for the 12 mm ball over a pressure range of 3 kPa to 50 kPa; this does not correspond closely to any of the predicted scaling relationships, however, it is closest to the high-speed regime, though it is ≈ 4 standard deviations higher than the predicted exponent value of $4/5 = 0.80$. Further experimental data is required to confirm whether the 12 mm ball exhibits the high-speed power law or not.

If the 12 mm ball does exhibit the high-speed power law, potential causes of the discrepancy in our measurements may include overfitting in the piecewise linear regression model or inaccurate anomaly detection by the RANSAC algorithm[31]. Thomas Sigsworth, who manually calculated the ball velocities, also found an exponent value slightly higher than 0.80 for the 12 mm ball, suggesting there may have been a systematic error in the experimental setup. Potential causes for systematic errors will be discussed in chapter 5.

4.1.4 13 mm

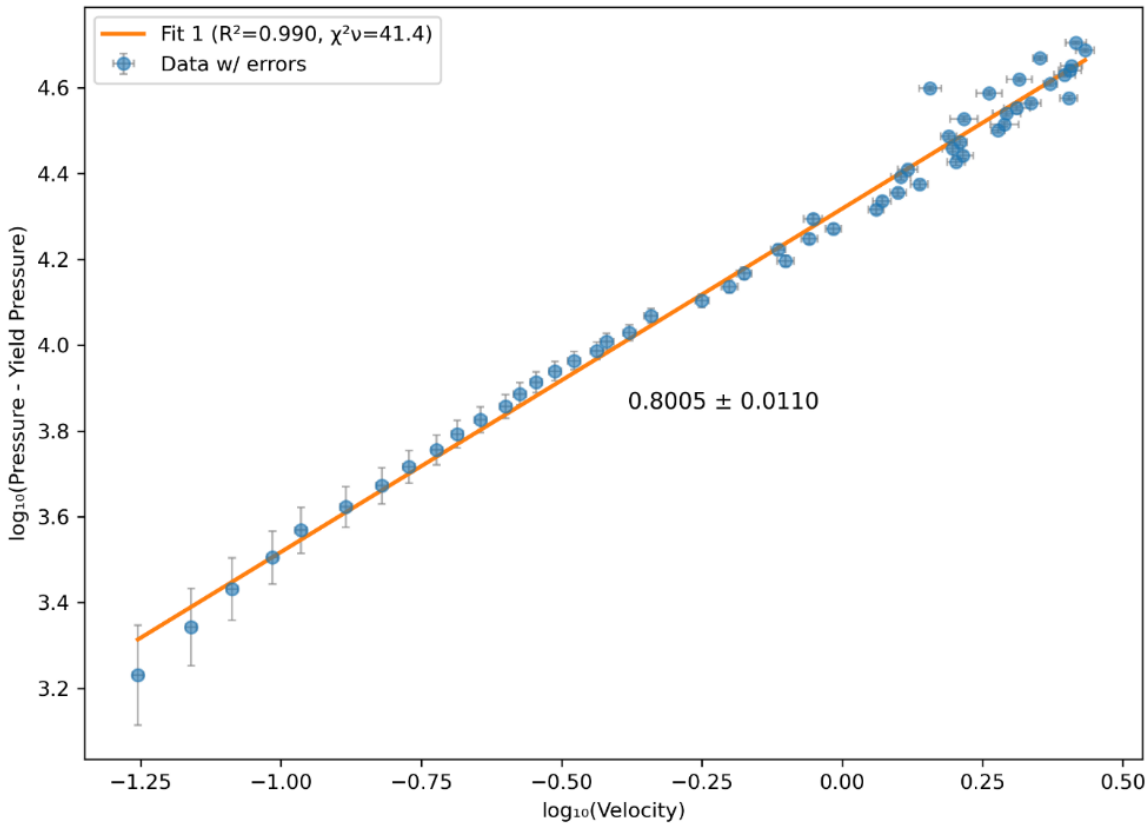

Figure 4.4: Pressure-Velocity plot for 13 mm diameter ball.

For the 13 mm ball, the value of the slenderness parameter is $s = 0.1813$ and radial excess parameter $\epsilon = 0.30$

Figure 4.4 indicates a power-law exponent of 0.8005 ± 0.0110 for the 13 mm ball over a pressure range of 2 kPa to 50 kPa, which corresponds to the high-speed power law[1].

4.1.5 14 mm

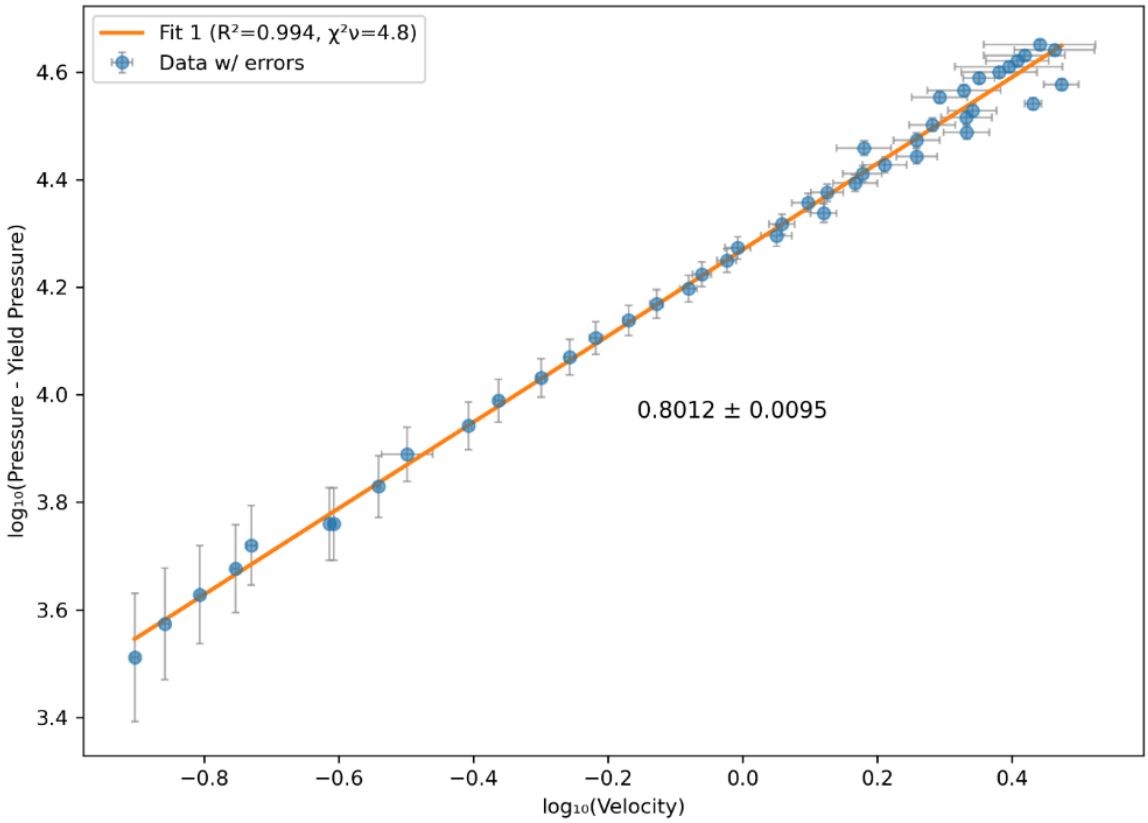

Figure 4.5: Pressure-Velocity plot for 14 mm diameter ball.

For the 14 mm ball, the value of the slenderness parameter is $s = 0.2047$ and radial excess parameter $\epsilon = 0.40$

Figure 4.5 indicates a power-law exponent of 0.8012 ± 0.0095 for the 14 mm ball over a pressure range of 3 kPa to 45 kPa, which corresponds to the high-speed power law[1].

4.1.6 15 mm

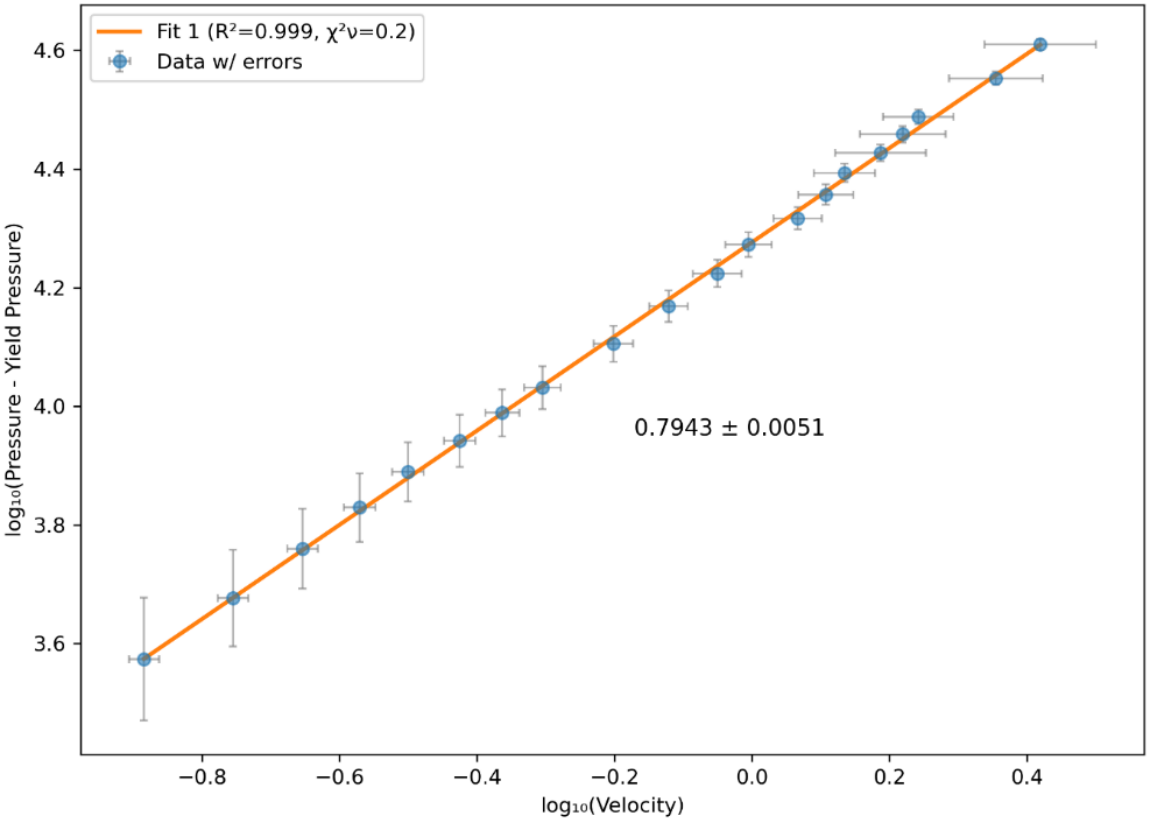

Figure 4.6: Pressure-Velocity plot for 15 mm diameter ball.

For the 15 mm ball, the value of the slenderness parameter is $s = 0.2241$ and radial excess parameter $\epsilon = 0.50$

Figure 4.6 indicates a power-law exponent of 0.7943 ± 0.0051 for the 15 mm ball over a pressure range of 4 kPa to 40 kPa, which corresponds to the high-speed power law[1]. The power law exponent is within 2 standard deviations (95% confidence interval) of the predicted value of $4/5 = 0.80$.

So far, we have been able to reliably observe the $\lambda \gg 1$ regime in the 11 mm, 13 mm, 14 mm, and 15 mm ball experiments. In comparison with the high-speed regime observed for the 11 mm ball, subsequent experiments have shown the same scaling behaviour over pressure ranges approximately 4 times larger and have consistently exhibited smaller measurement uncertainties. These observations lend support to the validity of the nonlinear membrane tension theory[1].

4.1.7 18 mm

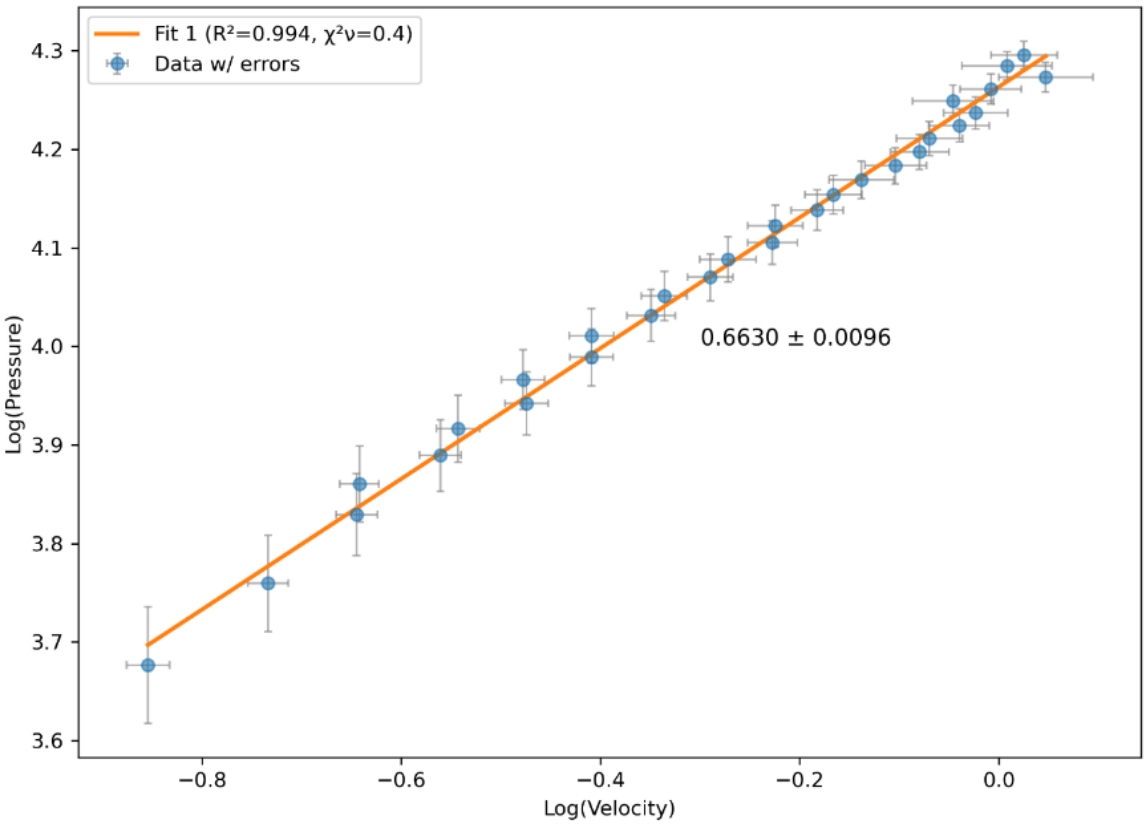

Figure 4.7: Pressure-Velocity plot for 18 mm diameter ball.

For the 18 mm ball, the value of the slenderness parameter is $s = 0.2677$ and radial excess parameter $\epsilon = 0.80$

Figure 4.7 indicates a power-law exponent of 0.6630 ± 0.0096 for the 18 mm ball over a pressure range of 5 kPa to 20 kPa, which corresponds to the intermediate speed $V^{2/3}$ power law (equation 2.10)[1]. The exponent value is within 1 standard deviation of the predicted value of $2/3 = 0.\overline{6}$; however, the pressure range does not span an entire order of magnitude.

The experiment was stopped at 37.5 kPa due to oil leakage from the tube, whereas the previous experiments were stopped between 50 kPa and 60 kPa. We could have observed this scaling law over a broader pressure range or a transition boundary between the intermediate and high-speed regimes, were it not for the apparatus's limits. Future experiments should maximise the structural integrity of the apparatus at higher pressures to allow for the observation of the full range of scaling laws.

4.2 Dimensionless scaling

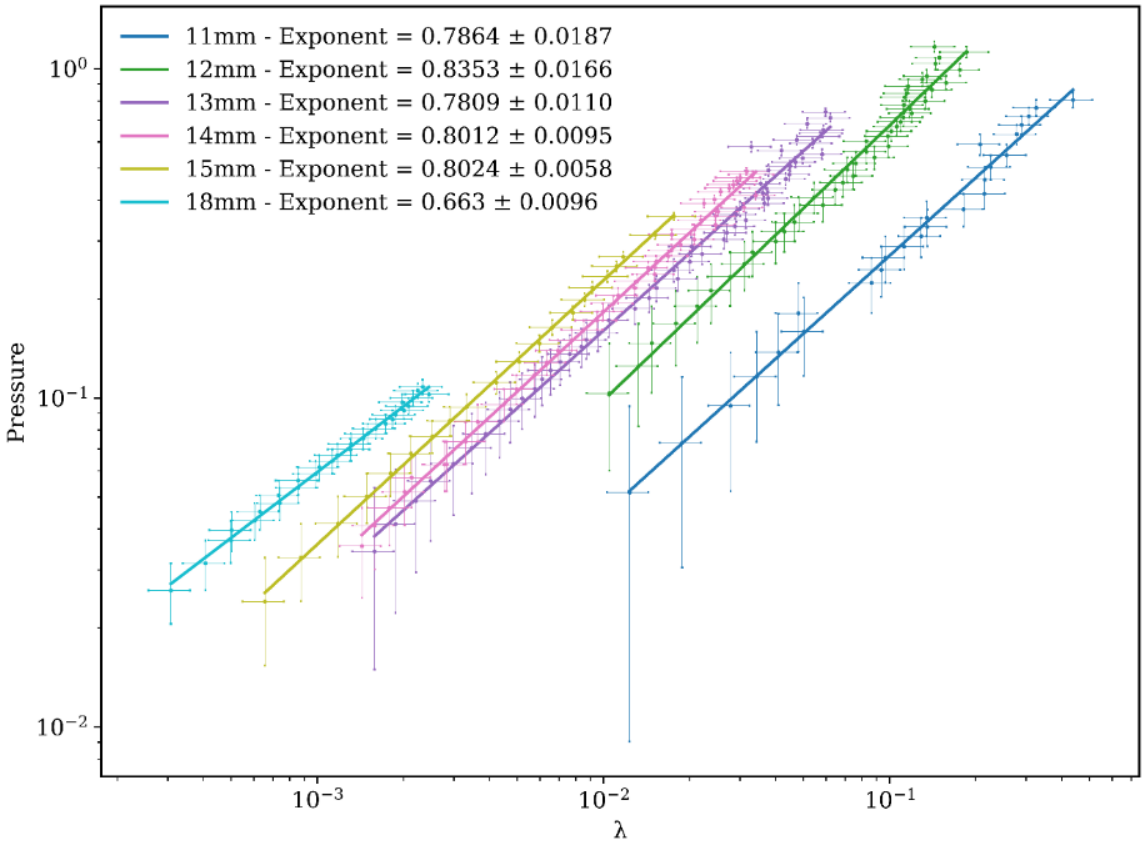

Figure 4.8: Dimensionless pressure-lambda plot for balls through 11 mm to 18 mm.

Figure 4.8 shows similar power law exponents to the pressure vs velocity linear fits, and allows for a comparison of the datasets in the non-dimensional λ parameter space. The difference in the exponents between the non-dimensional and dimensional analyses may be attributed to the propagation of uncertainties from the rescaling factors: Young's modulus, tube radius, tube wall thickness, and maximum radial excess.

To compare pressure more directly with the non-dimensional drag force, a dimensionless pressure parameter was defined as the drag force rescaling factor divided by the lubrication film area. We approximated the area over which the thin film exerts pressure to be

$$A \approx 2\pi a \delta_m \tag{4.1}$$

where a is the ball radius and δ_m is the maximum radial excess.

Moreover, dividing the drag force rescaling factor by the thin film area gives the pressure rescaling factor:

$$\mathcal{P} \simeq \frac{\mathcal{F}}{A} \simeq \frac{2\pi E b \delta_m^2}{a} \cdot \frac{1}{2\pi a \delta_m} \simeq \frac{E b \delta_m}{a^2} \tag{4.2}$$

This pressure rescaling coincides with the hoop stress resistance of the tube wall:

$$\sigma_h \propto \frac{E b u_r(z)}{a^2} \tag{4.3}$$

where $u_r(z)$ is the radial excess along the longitudinal axis of the intruder[1,37,38], therefore, when the non-dimensional pressure equals 1, it equals the hoop stress resistance.

The results for the 10 mm ball were omitted from Figure 4.8 because its maximum radial excess, $\delta_m = 0.01 \pm 0.161$ mm, is significantly smaller than the measurement uncertainty, which is obtained by quadrature from the uncertainties of the tube radius (±0.16 mm) and ball diameter (±0.02 mm). Therefore, reliable calculations for λ cannot be made for this ball.

Additionally, the linear scaling of pressure and velocity for the 10 mm ball implies that the maximum radial excess is $\delta_m = 0$, for which λ is ill-defined.

Chapter 5

Discussion

Our results provide encouraging evidence for the high-speed and intermediate power laws at work in a pressure-driven system analogous to many real-world applications, such as the movement of blood cells through capillaries. This strongly supports the theoretical approach outlined by Rallabandi et al.[1] and shows that it is likely a good model for the dynamics of rigid, pressure-driven spheres in narrow, fluid-filled elastic tubes.

5.1 Methodological limitations

In this section, we outline the limitations of our experimental method, which may have contributed to systematic errors in our results.

An unexpected but frequently occurring phenomenon we encountered was 'stick-slipping'[16], where, at low pressures, the ball would move a small distance through the tube, suddenly become stuck, then, after a small delay (or sometimes requiring a gentle push), move a small distance again. According to a 2022 paper by Zakhari and Bonnecaze (on soft permeable particles)[16], this is likely because, at low speeds, the pressure in the lubrication film around the ball drops very low, causing the particle to make contact with and stick to the tube wall.

Moreover, we occasionally experienced delays of up to 30 minutes between successive trials. During these intervals, the ball remained stationary within the tube, allowing the tube walls to compress the lubricant film around it. This may have thinned the lubrication layer, exacerbating stick-slip motion and increasing the apparent yield pressure. Although the dynamic film thickness recovers once flow resumes, transient variations in film thickness may have introduced random errors in yield pressure measurements.

Furthermore, oil absorption into the silicone may alter its elastic modulus over multi-hour or multi-day experiments[35]. While our small-strain Young's modulus measurements revealed no statistically significant difference between "dry" and "wet" specimens, the limited sample size warrants a more thorough investigation. Potential changes in the tube's elastic behaviour

could affect the validity of the scaling laws and introduce errors in the λ calculations.

Additionally, the ball was held stationary by hand at the inlet during each trial while the pump was pressurised to the target value. This often caused the silicone tube to bulge around the stationary ball due to the backpressure buildup. Upon release, the ball may accelerate rapidly in a bulged section of the tube, thereby displaying motion outside the deformation regime. Although this may initially produce a local velocity overshoot, its influence is confined to the first few centimetres; thereafter, the tube rapidly recoils to a diameter smaller than the ball, and the ball moves within the deformation regime. We are confident this effect is negligible in our results due to the RANSAC algorithm[31,39] filtering out the initial transient motion and other anomalies, but future work may benefit from investigating this in more depth to confirm that it does not influence the results in a pressure-driven system.

Another potential source of systematic error in our experimental procedure is parallax and frame distortion introduced by the ultra-wide-angle lens used in our video recording setup. The 6.0 mm focal length lens has a very wide field of view and may cause distortion around the edges of the frame or cause the centre of the frame to appear slightly magnified. This could lead to the propagation of systematic errors in the velocimetry data analysis. However, due to the RANSAC algorithm[31,39] heavily filtering out positional outliers at both the beginning and end of the position-time series, we believe this effect to be negligible.

5.2 Comparison in λ parameter space

The results show that the values of λ decrease significantly as the ball diameter increases, resulting in power laws occurring at λ values 2-3 orders of magnitude lower than those expected from the direct numerical analysis.

The cause may be the significant axial stretching we observed at high pressures. At pressures above 30 kPa, we observed the silicone tube stretching significantly along the axis of the tube, which presented itself as bending in a 'snake-like' action, which was likely caused by the fact that the tube was fixed at both ends, so axial stretching would result in lateral bending. This stretching may alter the hoop stress resistance[40,41], axial membrane tension, and lubrication film geometry relative to theory, and provides a possible explanation for our observations of the intermediate- and high-speed scaling laws at λ values significantly lower than those predicted by theory.

Our experimental results lend empirical weight to the theoretical work of Rallabandi et al.[1] by revealing the intermediate and high-speed power-law regimes. The absence of the low-speed regime likely reflects the minimum speeds accessible in our pressure-driven setup, which may have yeidl pressures that exceed the transition threshold for the low-speed regime.

Furthermore, in our experiment, the slender intruder condition ($s \ll 1$) is weakly fulfilled at larger sphere diameters. Rallabandi et al. state' the theory provides good approximations

of the numerical results for small λ up to $\varepsilon \approx 0.4$[1]. In principle, spheres are considered slender and will follow the same scaling laws as ellipsoids, and the intermediate and high-speed scaling laws are derived from direct numerical simulations, which provide a good model for the drag force for a wide range of intruder shapes over the entire λ parameter space[1].

However, we are concerned that the 15 mm and 18 mm balls cross this threshold and may not strictly behave like slender intruders. This may result in the intruders crossing into the intermediate and high-speed regimes at lower values of λ than an ellipsoid intruder.

Rallabandi et al. highlight this possibility by indicating the conditions required to observe the different scaling laws may differ for an ellipsoid intruder depending on which of its sides are forward-facing[1]. In other words, an egg-shaped intruder in an elastic tube will move differently depending on whether the pointed or rounded end is facing forward. Our large spheres may be analogous to the rounded end of the egg shape as they both have large dimensionless curvature values (equation 2.9).

Therefore, to minimise differences with the theory and to seek the scaling laws in the predicted regions of the λ parameter space, our experiments should be repeated with ellipsoidal intruders.

5.3 Future work

Future experiments should employ tubes spanning a wider Young's modulus range and a larger range of viscous lubricants to probe the asymptotic limits of the theory. Highly viscous lubricants could slow motion sufficiently to resolve transition thresholds and validate the $V^{1/2}$ scaling of the low-speed regime. Meanwhile, more rigid tube materials, such as PVC, may exhibit behaviour when membrane deformation is small but non-zero.

Another benefit of using a tube material with a higher Young's modulus would be the potential reduction of unwanted axial stretching at very high pressures, which we hypothesise is the main cause of the intermediate and high-speed scaling laws occurring at lower values of λ than predicted by the theory.

Additionally, using a direct-drive apparatus which exerts a mechanical force to move the ball at a constant velocity through the tube (as is being done by the ESPCI group in Paris[42]) would measure the drag force directly rather than infer it from pressure and eliminate uncertainties from Poiseuille flow and hydrostatic pressure drops in the tube. A directly-driven intruder is more similar to the theoretical basis set by Rallabandi et al. than a pressure-driven system, as they describe the motion as if the intruder is 'pulled by a string'[1]. This would also allow the experiments to be conducted at significantly lower speeds and pressures, potentially enabling the low-speed regime or regime transition boundaries to be observed.

Similarly, reducing the volume of lubricant surrounding the intruder would further

suppress flow contributions and simplify assumptions about pressure losses. Combining these changes would isolate the lubrication pressure, hoop stress resistance, and axial membrane tension, thereby providing more reliable results.

Finally, beyond the direct extensions of this experiment, it would be of great interest to devise an experiment that analyses the exact shape of the thin film lubrication layer and measures its thickness along the axial length of the intruder. Measuring the lubrication thickness as a function of translational velocity would provide even deeper insight into the elastohydrodynamics of this system.

Chapter 6

Conclusion

This report has outlined the experimental investigation of the motion of rigid, oversized spherical intruders through a fluid-filled elastic tube. Theoretical analyses of this system have progressed significantly from Lighthill's 1968 analysis[19] to the inclusion of axial membrane tension by Rallabandi et al. in 2021[1]. This system has many analogues in real-world applications, including red blood cells in capillaries, eggs in oviducts, soft robotics, microfluidic medical devices, and more. We sought to investigate this system experimentally and compare it with the state-of-the-art theoretical predictions.

We designed and built an experimental apparatus that drove spherical intruders through a silicone tube filled with rapeseed oil. The intruder was driven through the tube by applying a hydraulic pressure to one of the connected fluid reservoirs, and the ball's motion was recorded using a high-framerate camera. The experimental data were analysed using a custom Python script to calculate the ball's speed from the pressure. Then, a piecewise linear fit was performed to identify potential power-law relationships between pressure and velocity.

We observed the high-speed scaling relationship (equation 2.11) for the 11 mm, 13 mm, 14 mm, and 15 mm balls. Meanwhile, we observed the intermediate speed relationship (equation 2.10) for the 18 mm ball. We observed a linear scaling for the 10 mm ball, likely due to negligible tube deformation. For the 12 mm ball, we observed a power law with an exponent that may indicate the high-speed regime; however, further experiments are needed to investigate this possibility. We did not observe the low-speed scaling relationship (equation 2.8) in any of our experiments.

Our results provide empirical support for the existence of intermediate- and high-speed power laws, as predicted by Rallabandi et al.[1]. However, due to various methodological limitations, we recommend conducting further experiments with ellipsoid intruders. Further investigations should also implement an experimental procedure that accounts for the mechanical movement of the intruder through the tube to mitigate pressure-driven flow effects. Implementing these recommendations will strengthen the bridge between theory and experiment, yielding a more comprehensive validation of elastohydrodynamic scaling laws.

Bibliography

[1] B. Rallabandi, J. Eggers, M. A. Herrada, and H. A. Stone, "Motion of a tightly fitting axisymmetric object through a lubricated elastic tube", Journal of Fluid Mechanics **926**, A27 (2021).

[2] J. M. Barakat and E. S. G. Shaqfeh, "The steady motion of a closely fitting vesicle in a tube", Journal of Fluid Mechanics **835**, 721 (2018).

[3] J. R. G. Bradfield, "Radiographic studies on the formation of the hen's egg shell", The Journal of Experimental Biology **28**, 125 (1951).

[4] A. Salamon and J. P. Kent, "Orientation of the Egg at Laying - Is the Pointed or the Blunt End First?", International Journal of Poultry Science **13**, 316 (2014).

[5] D. G. Blackburn, "Structure, function, and evolution of the oviducts of squamate reptiles, with special reference to viviparity and placentation", Journal of Experimental Zoology **282**, 560 (1998).

[6] P. W. Wilson, C. S. Suther, M. M. Bain, et al., "Understanding avian egg cuticle formation in the oviduct: A study of its origin and deposition", Biology of Reproduction **97**, 39 (2017).

[7] K. E. Barrett, S. M. Barman, H. L. Brooks, and J. X.-J. Yuan, in *Ganong's Review of Medical Physiology, 27th Edition*, Book, Section vols. (McGraw Hill, New York, NY, 2025).

[8] J. Freund, "Numerical Simulation of Flowing Blood Cells", Annual Review of Fluid Mechanics **46**, `10.1146/annurev-fluid-010313-141349` (2013).

[9] P. G. Vann and J. M. Fitz-Gerald, "Flow mechanics of red cell trains in very narrow capillaries. I. Trains of uniform cells", Microvascular Research **24**, 296 (1982).

[10] J. M. Fitz-Gerald, "Mechanics of red-cell motion through very narrow capillaries", Proceedings of the Royal Society B **174**, 193 (1969).

[11] T. W. Secomb, R. Skalak, N. Özkaya, and J. F. Gross, "Flow of axisymmetric red blood cells in narrow capillaries", Journal of Fluid Mechanics **163**, 405 (1986).

[12] T. M. Squires and S. R. Quake, "Microfluidics: Fluid physics at the nanoliter scale", Reviews of Modern Physics **77**, 977 (2005).

[13] D. J. Laser and J. G. Santiago, "A review of micropumps", Journal of Micromechanics and Microengineering **14**, R35 (2004).

[14] A. Grüntzig, "Transluminal dilatation of coronary-artery stenosis", The Lancet **1**, 263 (1978).

[15] A. Vurgaft, S. B. Elbaz, and A. D. Gat, "Forced motion of a cylinder within a liquid-filled elastic tube – a model of minimally invasive medical procedures", Journal of Fluid Mechanics **881**, 1048 (2019).

[16] M. E. A. Zakhari and R. T. Bonnecaze, "Slip-stick transitions of soft permeable particles near a repulsive wall", Soft Matter **18**, 4239 (2022).

[17] Bridgestone Corporation, *Development of a Hydraulic Drive High-Power Artificial Muscle through the Cabinet Office Tough Robotics Challenge News | Bridgestone*, Bridgestone Global Website, (Jan. 26, 2017) `https://www.bridgestone.com/corporate/news/2017012601.html.html` (visited on 05/02/2025).

[18] B. Saintyves, T. Jules, T. Salez, and L. Mahadevan, "Self-sustained lift and low friction via soft lubrication", Proceedings of the National Academy of Sciences **113**, 5847 (2016).

[19] M. J. Lighthill, "Pressure-forcing of tightly fitting pellets along fluid-filled elastic tubes", Journal of Fluid Mechanics **34**, 113 (1968).

[20] M. Tani, T. Cambau, J. Bico, and E. Reyssat, "Motion of a rigid sphere through an elastic tube with a lubrication film", in , Vol. 2017 (Mar. 1, 2017), p. C15.013.

[21] B. Rallabandi, J. Marthelot, É. Jambon-Puillet, P.-T. Brun, and J. Eggers, "Curvature regularisation near contacts with stretched elastic tubes", Physical Review Letters **123**, 168002 (2019).

[22] P. M. Bungay and H. Brenner, "The motion of a closely-fitting sphere in a fluid-filled tube", International Journal of Multiphase Flow **1**, 25 (1973).

[23] Y.-C. Li, Y.-C. Liao, T.-C. Wen, and H.-H. Wei, "Breakdown of the Bretherton law due to wall slippage", Journal of Fluid Mechanics **741**, 200 (2014).

[24] A. Juel and A. Heap, "The reopening of a collapsed fluid-filled elastic tube", Journal of Fluid Mechanics **572**, 287 (2007).

[25] Elveflow, *Elveflow ESI User Guide*, Elveflow Knowledge Center, `https://support.elveflow.com/support/solutions/articles/48000984977-user-guide-of-the-elveflow-smart-interface-esi-` (visited on 04/17/2025).

[26] Elveflow, *User Guide OB1 Mk3+*, Elveflow Knowledge Center, `https://support.elveflow.com/support/solutions/articles/48001225594-user-guide-ob1-mk3-` (visited on 04/17/2025).

[27] JAI, *eBUS SDK Older Versions*, JAI, https://www.jai.com/support-software/older-versions/ (visited on 03/14/2025).

[28] T. Ferreira and W. Rasband, "ImageJ User Guide",

[29] S. Developers, *Find_peaks — SciPy v1.15.0 Manual*, https://docs.scipy.org/doc/scipy-1.15.0/reference/generated/scipy.signal.find_peaks.html (visited on 04/26/2025).

[30] N. Developers, *Numpy.polyfit — NumPy v2.2 Manual*, https://numpy.org/doc/2.2/reference/generated/numpy.polyfit.html (visited on 04/17/2025).

[31] scikit-learn developers, *RANSACRegressor*, scikit-learn, https://scikit-learn/stable/modules/generated/sklearn.linear_model.RANSACRegressor.html (visited on 04/17/2025).

[32] C. F. Jekel and G. Venter, *Pwlf: A Python library for fitting 1D continuous piecewise linear functions*, 2019.

[33] Anton Paar, *Modular Compact Rheometer: MCR*, Anton Paar, https://www.anton-paar.com/uk-en/products/details/modular-compact-rheometer/ (visited on 04/24/2025).

[34] Instron, *Universal Testing Systems | Instron*, (Sept. 3, 2024) https://www.instron.com/en/products/testing-systems/universal-testing-systems/ (visited on 04/19/2026).

[35] A. Mahomed, D. W. L. Hukins, and S. N. Kukureka, "Swelling of medical grade silicones in liquids and calculation of their cross-link densities", Medical Engineering & Physics **32**, 298 (2010).

[36] A. E. Vardy, M. I. G. Bloor, and J. A. Fox, "Capsular flow in pipelines", Journal of Fluid Mechanics **56**, 49 (1972).

[37] W. T. Koiter, "On the nonlinear theory of thin elastic shells", Proceedings of the Royal Netherlands Academy of Arts and Sciences, Series B **69**, 1 (1966).

[38] L. D. Landau and E. M. Lifshitz, *Theory of elasticity*, 3rd (Pergamon, 1986).

[39] M. A. Fischler and R. C. Bolles, "Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography", Commun. ACM **24**, 381 (1981).

[40] A. P. Boresi and R. J. Schmidt, *Advanced Mechanics of Materials* (John Wiley & Sons, Oct. 22, 2002), 715 pp.

[41] S. Timoshenko, *Theory of elasticity*, 3rd ed., Engineering Societies Monographs (McGraw Hill, New York ; 1970).

[42] M. Tani, T. Cambau, É. Reyssat, and J. Bico, *Motion of a sphere confined in a flexible lubricated tube*, Technical Report (PMMH-ESPCI-PSL, UMR CNRS 6738, UPMC, Paris, France, Sept. 2024).

Appendix A

Table of Results

Ball Diameter (mm)	Velocity Exponent	σ_V	Pressure Range (kPa)
10	0.9800	± 0.0286	1.0 - 40.0
11	0.8050	± 0.0189	1.0 - 16.0
12	0.8570	± 0.0139	3.0 - 50.0
13	0.8005	± 0.0110	2.0 - 50.0
14	0.8012	± 0.0095	3.0 - 45.0
15	0.7943	± 0.0051	4.0 - 40.0
18	0.6630	± 0.0096	5.0 - 20.0

Table A.1: Table of velocity results

Ball Diameter (mm)	λ Exponent	σ_λ	Pressure Range (kPa)
11	0.7864	± 0.0187	1.0 - 16.0
12	0.8353	± 0.0166	3.0 - 50.0
13	0.7809	± 0.0110	2.0 - 50.0
14	0.8012	± 0.0095	3.0 - 45.0
15	0.8024	± 0.0058	4.0 - 40.0
18	0.6630	± 0.0096	5.0 - 20.0

Table A.2: Table of λ results

Table A.2 shows the power law exponents for the non-dimensional pressure vs. lambda fits, alongside the pressure ranges over which the data is fitted.

Appendix B

Images

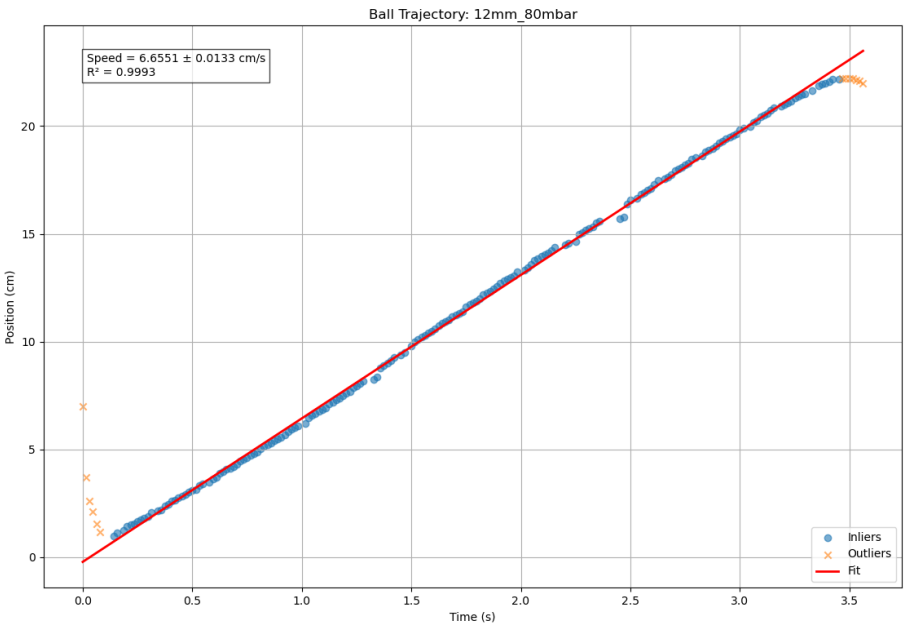

Figure B.1: Position-time plot for 12 mm ball at 80 mbar pressure

In figure B.1 you can see the outliers (orange crosses) that have been identified by the RANSAC algorithm and have been filtered out of the data before the final velocity is calculated.

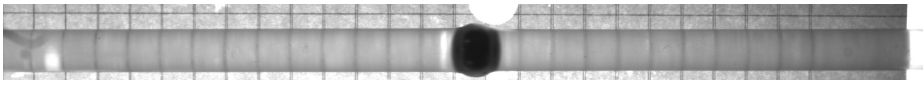

Figure B.2: A frame from the video of the 12 mm ball falling at 80 mbar pressure

Figure B.2 shows the ball in the centre of the tube, approximately halfway through its trajectory. The background contains an 8 mm lined paper that was used for pixel calibration. The ball is illuminated by the LED lights positioned below the lined paper.

Appendix C

Code

C.1 Velocimetry

```python
import os
import glob
import logging

import numpy as np
from scipy.stats import linregress
from scipy.signal import savgol_filter
from PIL import Image
import matplotlib.pyplot as plt

# Optional robust regression
try:
from sklearn.linear_model import RANSACRegressor, LinearRegression

SKLEARN_AVAILABLE = True
except ImportError:
SKLEARN_AVAILABLE = False
logging.warning(
"scikit-learn not available: robust regression will be disabled."
)

# Configure logging
logging.basicConfig(level=logging.INFO, format="%(levelname)s: %(message)s")

# ===== User-configurable settings =====
# Set the folder containing TIFF frames:
FOLDER_PATH = "10mm_440mbar"
# Pixel-to-cm conversion factor:
PIXEL_TO_CM = 108.3
# Adjust STDEV multiplier for outliers (lower is more strict)
STRICTNESS = 1
```

```python
# Toggle robust regression (RANSAC outlier filtering):
USE_ROBUST = True
# RANSAC residual threshold (in cm); set to None to use default 0.5 * std:
RANSAC_THRESHOLD = STRICTNESS
# Temporal smoothing settings (Savitzky-Golay):
SMOOTH_WINDOW = 5  # window length (odd integer >= 3)
SMOOTH_POLYORDER = 2  # polynomial order ( < SMOOTH_WINDOW)

def extract_epoch_from_filename(filename: str) -> int:
"""
Extracts the hexadecimal timestamp from 'prefix_<hex>.<ext>' and converts to int.
"""
base = os.path.splitext(os.path.basename(filename))[0]
if "_" in base:
_, hex_part = base.split("_", 1)
try:
return int(hex_part, 16)
except ValueError:
logging.warning(f"Invalid hex in filename: {base}")
return None

def process_images(folder_path: str) -> list[tuple[str, float, None, int]]:
"""
Sorts frames by epoch; loads, inverts, and locates the ball via 2D
↪  intensity-weighted centroid.
Returns list of (file_path, x_centroid, None, epoch).
"""
paths = glob.glob(os.path.join(folder_path, "*.tif")) + glob.glob(
os.path.join(folder_path, "*.tiff")
)
valid = []
for p in paths:
e = extract_epoch_from_filename(p)
if e is not None:
valid.append((p, e))
else:
logging.debug(f"Skipping {p} (no epoch)")
valid.sort(key=lambda x: x[1])

results = []
for path, epoch in valid:
img = Image.open(path).convert("L")
arr = np.array(img, dtype=float)
inv = 255.0 - arr

# 2D intensity-weighted centroid detection
mean_val = inv.mean()
std_val = inv.std()
```

```
thresh = mean_val + 1.96 * std_val
mask = inv > thresh

if np.any(mask):
y_idx, x_idx = np.nonzero(mask)
values = inv[y_idx, x_idx]
x_centroid = float(np.sum(x_idx * values) / np.sum(values))
else:
logging.warning(f"No bright region detected in {path}")
x_centroid = None

results.append((path, x_centroid, None, epoch))
return results

def calculate_speed_fit(
detections: list[tuple[str, float, None, int]],
pixel_to_cm: float,
robust: bool,
ransac_residual_threshold: float,
) -> tuple[list[float], list[float], dict, np.ndarray]:
"""
Computes linear regression of position vs time; applies temporal smoothing and
↪  optional RANSAC.
Returns times (s), raw positions (cm), fit stats, and boolean inlier_mask.
Fit stats include slope, intercept, r_value, p_value, stderr.
"""
epochs = [e for *_, e in detections if e is not None]
if not epochs:
raise ValueError("No valid epochs for time.")
t0 = min(epochs)

times, positions = [], []
for _, x, *_, e in detections:
if x is not None and e is not None:
times.append((e - t0) * 1e-3)
positions.append(x / pixel_to_cm)

if len(times) < 2:
raise ValueError("Not enough data points for regression.")

times_arr = np.array(times)
pos_arr = np.array(positions)

# Apply temporal smoothing if enough points
window = SMOOTH_WINDOW
if len(pos_arr) >= window and window % 2 == 1:
pos_smooth = savgol_filter(
pos_arr, window_length=window, polyorder=SMOOTH_POLYORDER
```

```
)
else:
pos_smooth = pos_arr

# Choose array for regression
X = times_arr.reshape(-1, 1)
y = pos_smooth

# Robust regression with RANSAC
if robust and SKLEARN_AVAILABLE:
base = LinearRegression()
thresh = (
ransac_residual_threshold
if ransac_residual_threshold is not None
else np.std(y) * STRICTNESS
)
ransac = RANSACRegressor(
estimator=base, residual_threshold=thresh, random_state=0
)
ransac.fit(X, y)
inliers = ransac.inlier_mask_
# Refit OLS on inliers for statistics
lr = linregress(times_arr[inliers], y[inliers])
else:
inliers = np.ones_like(times_arr, dtype=bool)
lr = linregress(times_arr, y)

fit = {
        "slope": lr.slope,
        "intercept": lr.intercept,
        "r_value": lr.rvalue,
        "p_value": lr.pvalue,
        "stderr": lr.stderr,
    }
return times, positions, fit, inliers

def plot_speed_fit(
times: list[float],
positions: list[float],
fit: dict,
folder_name: str,
inliers: np.ndarray = None,
) -> None:
"""
Plots original positions vs time, regression fit, and highlights
↪  inliers/outliers.
Saves plot to 'results_{folder_name}.png'.
"""
```

```
plt.figure(figsize=(8, 6))
t_arr = np.array(times)
p_arr = np.array(positions)
if inliers is not None:
plt.scatter(
t_arr[inliers], p_arr[inliers], label="Inliers", marker="o"
)
plt.scatter(
t_arr[~inliers], p_arr[~inliers], label="Outliers", marker="x"
)
else:
plt.scatter(times, positions, label="Data", marker="o")

# Plot fit line
t_fit = np.linspace(t_arr.min(), t_arr.max(), 200)
y_fit = fit["slope"] * t_fit + fit["intercept"]
plt.plot(t_fit, y_fit, label="Fit", lw=2)

plt.xlabel("Time (s)")
plt.ylabel("Horizontal Position (cm)")
plt.title(f"Ball Trajectory for {folder_name}")
annot = (
f"Speed = {fit['slope']:.4f} ± {fit['stderr']:.4f} cm/s\n"
f"R² = {fit['r_value']**2:.4f}"
)
plt.text(
0.05,
0.95,
annot,
transform=plt.gca().transAxes,
va="top",
bbox=dict(facecolor="white", alpha=0.7),
)
plt.legend()
plt.grid(True)

out_fname = f"results_{folder_name}.png"
plt.savefig(out_fname, dpi=300)
logging.info(f"Plot saved to {out_fname}")
plt.show()

print(f"{fit['slope']:.4f}")
print(f"{fit['stderr']:.4f}")

if __name__ == "__main__":
# Process images and compute fit using configured settings
detections = process_images(FOLDER_PATH)
times, positions, fit_info, inliers = calculate_speed_fit(
```

```
detections,
pixel_to_cm=PIXEL_TO_CM,
robust=USE_ROBUST,
ransac_residual_threshold=RANSAC_THRESHOLD,
)
plot_speed_fit(
times, positions, fit_info, os.path.basename(FOLDER_PATH), inliers)
```

C.2 Piecewise Linear Regression

```
import numpy as np
import pandas as pd
import pwlf
import matplotlib.pyplot as plt

# Load data (must include columns 'speed','pressure','v_err','p_err')
df = pd.read_csv("10mm.csv")
speed = df["speed"].values
pressure = df["pressure"].values
logV_err = df["v_err"].values
logP_err = df["p_err"].values
ball_size = "10mm Ball"
max_fits = 1

# Transform to log10
logV = np.log10(speed)
logP = np.log10(pressure)

# Weights for weighted least squares
weights = 1.0 / logP_err

def find_optimal_segments(x, y, w, max_segments):
bic_list = []
models = []
n = len(x)
for k in range(1, max_segments + 1):
model = pwlf.PiecewiseLinFit(x, y)
breaks = model.fit(k)
sse = model.ssr
p = 2 * k
bic = n * np.log(sse / n) + p * np.log(n)
bic_list.append(bic)
models.append((model, breaks))
best = bic_list.index(min(bic_list))
return best + 1, models[best]


```

```python
# Fit piecewise model with weights
optimal_k, (model, breaks) = find_optimal_segments(
logV, logP, weights, max_fits
)

print(f"Optimal number of segments: {optimal_k}")
print("Breakpoints (log10 speed):", breaks)

# Slopes ± SE
slopes = model.calc_slopes()
ses_full = model.standard_errors()
slope_ses = ses_full[1:]
for i, (m, se) in enumerate(zip(slopes, slope_ses), start=1):
print(f" Segment {i} exponent: {m:.4f} ± {se:.4f}")

# Overall R²
print(f"Overall R²: {model.r_squared():.4f}")

# Full predictions
y_hat = model.predict(logV)

# Plot with error bars
plt.figure(figsize=(8, 6))
plt.errorbar(
logV,
logP,
xerr=logV_err,
yerr=logP_err,
fmt="o",
ecolor="grey",
elinewidth=1,
capsize=2,
alpha=0.6,
label="Data w/ errors",
)

# Per-segment metrics and plot
for i, (slope, se) in enumerate(zip(slopes, slope_ses)):
x0, x1 = breaks[i], breaks[i + 1]
mask = (logV >= x0) & (logV <= x1)
y_seg = logP[mask]
y_pred_seg = y_hat[mask]
sigma_seg = logP_err[mask]
n_i = mask.sum()

# R²
ss_res = np.sum((y_seg - y_pred_seg) ** 2)
ss_tot = np.sum((y_seg - y_seg.mean()) ** 2)
```

```python
r2 = 1 - ss_res / ss_tot

# reduced chi-squared
chi2_red = np.sum(((y_seg - y_pred_seg) / sigma_seg) ** 2) / (n_i - 1)

# Plot segment line
intercept = model.intercepts[i]
xs = np.linspace(x0, x1, 100)
ys = slope * xs + intercept
plt.plot(
xs,
ys,
linewidth=2,
label=f"Fit {i+1} (R square={r2:.3f}, reduced chi square={chi2_red:.1f})",
)

# annotate exponent ± SE
x_mid = 0.5 * (x0 + x1)
y_mid = slope * x_mid + intercept
plt.text(
x_mid+0.2,
y_mid-0.3,
f"{slope:.4f} ± {se:.4f}",
fontsize=11,
ha="center",
va="bottom",
)

plt.xlabel("Log(Speed [m/s])")
plt.ylabel("Log(Pressure [Pa])")
plt.title(f"Pressure vs Speed for {ball_size}")
plt.legend()
plt.tight_layout()
plt.savefig(f"Pressure vs Speed - Lower {ball_size} ({optimal_k} segments).png", dpi=300)
plt.show()
```